
Fig. 8: MindWave Mobile headset


Fig. 9: Neuro headset


Fig. 10: Muse headset


Fig. 11: Enobi headset

Fig. 12: BCI 2000

Fig. 13: Open Vibe

Fig. 14: AsTeRICS

TABLE II
OTHER IMPORTANT WORK DONE BY RESEARCHERS IN THE SIMILAR DOMAIN

Authors	Work done
Choaouachi and Frasson (2011)	Studied mental workload, engagement and emotions for ITS by using 6-channel EEG
Derbali and Frasson (2010)	Used EEG waves pattern to study players' motivation during different parts of a serious game
Kohlmorgen et al. (2010)	Developed real-time mental workload detection of drivers using EEG-based system
Heraz and Frasson (2007)	Predicted three major dimensions of the learner's emotions from brainwaves by using EEG for the development of an effective ITS
Berka et al. (2007)	Demonstrated that EEG correlates with task engagement and mental workload in vigilance, learning and memory tasks

Available hardware and software solutions

In recent years, a number of tools have been made available in the market that can be used to build such kind of systems. Details of these hardware and software solutions are given below.

Hardware solutions.

MindWave Mobile. Brainwave Starter Kit is a basic introduction to neuroscience and brainwave technology. It safely measures brainwave signals and monitors the attention levels of students as they interact with Maths, memory and pattern recognition applications. They have developed a number of mobile apps ranging from fun entertainment to serious learning like Meditation Journal, SpeedMath, BlinkZone, SpadeA, Find Number and MindHunter.

Epoc Neuro Headset. Emotiv EPOC is a high-resolution, neuro-signal acquisition, wireless headset. It utilises an arrangement of sensors to tune into electric signs delivered by the mind to identify players' thoughts, emotions and expressions, and connects remotely to most personal computers.

Muse. The brain-sensing headband helps to do more with the mind by providing exercises designed to manage stress and settle the mind.

Enobi. It is a wearable and wireless electrophysiology sensor system for recording EEG.

Software solutions.

BCI 2000. BCI 2000 is a general-purpose software system for brain-computer interface (BCI) research. It can also be used for data acquisition, stimulus presentation and brain-monitoring applications. BCI 2000 system is available for free for non-profit research and educational purposes.

OpenViBE. OpenViBE is a free and open source software platform for designing, testing and using BCI that addresses every type of user. It does not require any prerequisite knowledge and provides tools and applications for everybody.

AsTeRICS. This is a free and open source construction set for assistive technologies. It allows the creation of flexible solutions for people with disabilities using a large set of sensors and actuators.

AsTeRICS is an ongoing EU-funded project that brings together companies, foundations and universities with a single purpose. It is a flexible system adaptable to quickly develop assistive technologies applications.

Conclusion

When we watch 3D movies at a theatre, special goggles are given to us to have a better experience. The day is not far when special caps will be given to the learner that she or he will wear while learning online, and contents will be dynamically adjusted to support the current pace of learning of the learner.

Thus, the way we acquire knowledge and skills will fundamentally change in the near future due to the introduction of new technologies supporting flexible and personalised learning models. Cognitive science will play a transformational role in the world of education, and its adoption promises to offer effectiveness of intelligent tutoring at an unprecedented scale in the context of an open, collaborative, lifelong learning experience. EFY